

TrES-2b

R-1555

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_140813 · created 2026-08-01T20:26:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456882.84158 +/- 0.00032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.09 +/- 0.041
Transit depth (Rp/Rs)^2	0.81 +/- 0.73 [%]
Orbital Inclination (inc)	88.82 +/- 2.96
Airmass coefficient 1 (a1)	1.094 +/- 0.041
Airmass coefficient 2 (a2)	-0.054 +/- 0.023
Scatter in the residuals of the lightcurve fit is	3.84 %
Best Comparison Star	#3 - [304, 438]
Optimal Aperture	2.71
Optimal Annulus	12.46
Transit Duration (day)	0.099 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.09 ± 0.041 vs 0.1254 (deviation 0.86σ)
Duration fit vs published	0.099 d vs 0.0746 d (deviation 1.53σ)
Mid-time O–C	-0.2 min
Depth SNR	2.2
Residual scatter	3.84 %

Light curve

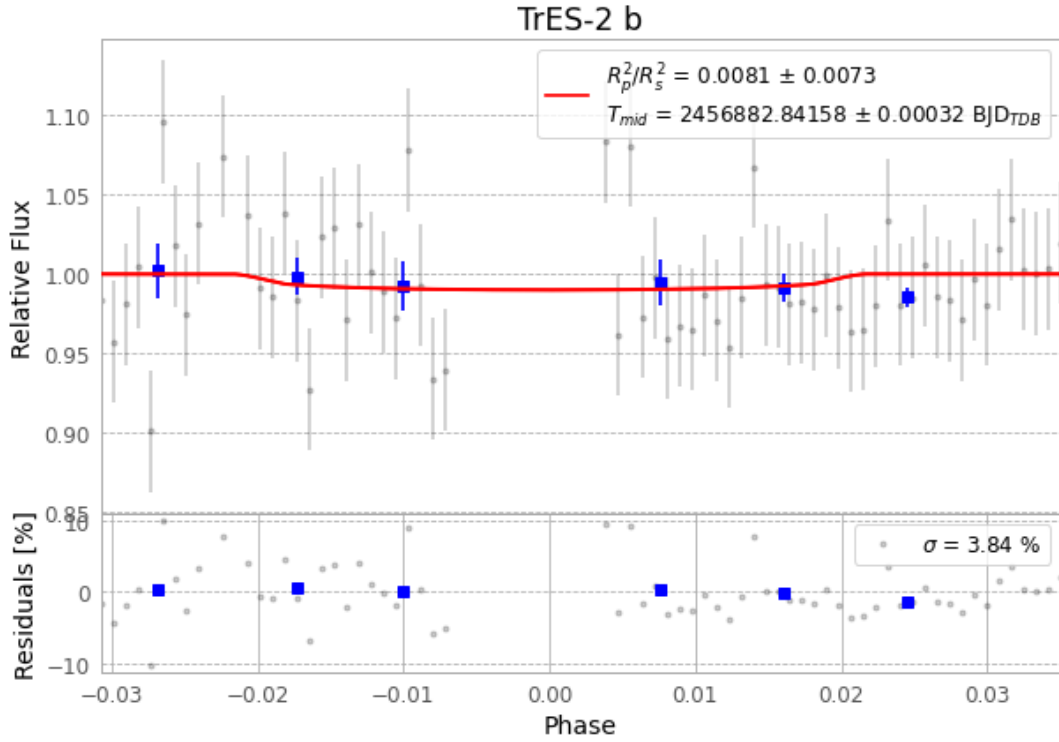


Figure 1 — FinalLightCurve_TrES-2 b_2014-08-12.png (SHA-256 2ec9d505381ea83e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 299], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ea3210eb511ad509...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), TRES-2140813061814.FITS → TRES-2140813101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-140813.log (087c8e253446...), FinalLightCurve_TrES-2 b_2014-08-12.png (2ec9d505381e...), FinalLightCurve_TrES-2 b_2014-08-12.csv (20a210cbc859...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 20:26Z.